



LORENTZ CENTER WORKSHOP

Accelerating Astrophysical Discovery

with Foundation Models

Participant booklet

14–18 September 2026 · Leiden, The Netherlands

WELCOME

A week built for making

Dear Lorentz Workshop participant,

In a little over a month, we will meet in Leiden for an interactive week devoted to pushing the bounds of how we study the cosmos. We want the workshop to turn diverse expertise into shared decisions, concrete artifacts, and collaborations that continue after Friday.

This booklet introduces the consortium's ambition, explains the four Key Problem Areas (KPAs), lays out the working timetable, and suggests a few things to think about before you arrive. You do not need polished proposals. Bring concrete examples, informed questions, and a willingness to build across disciplinary boundaries.

AT A GLANCE**Workshop**

Accelerating Astrophysical Discovery with Foundation Models

Dates

Monday 14 to Friday 18 September 2026

Venue

Lorentz Center@omega, Leiden, The Netherlands

Start times

Monday at 10:00; Tuesday–Friday at 9:30

The working style is closer to an open-source project hackathon than a sequence of lectures: short framing talks, focused working sessions, plenary synthesis, and a named deliverable each day.

THE CONSORTIUM

Two connected pillars

PILLAR 01**Design science for human–AI collaboration**

The first goal is to design how future astrophysicists and cosmologists will do science in collaboration with AI. This is more than adding chat interfaces to existing tools. It means redesigning workflows so humans, reasoning language models, simulators, archives, instruments, and foundation models can jointly form hypotheses, retrieve evidence, run checks, update beliefs, and communicate results.

PILLAR 02**Build a joint representation of physical data**

The second goal is to build the technical substrate: a model that learns across observations, simulations, synthetic observations, language, code, and instrumental context. It should support meaningful embeddings for retrieval and serendipity, along with generative capabilities for conditional inference across linked views of the same physical system.

One shared ambition. The two pillars meet in an agentic discovery system: reasoning models guide scientific work, the joint-representation model grounds that work in physical evidence, and humans remain responsible for judgement.

KEY PROBLEM AREAS

KPAs 1 and 2

The KPAs are not four separate projects. They are four tests of whether a shared model can understand astrophysical data well enough to accelerate discovery.

1 Serendipity through embeddings

A physically meaningful embedding space can surface rare objects, surprising analogues, counterexamples, and neighbouring evidence a scientist may not know to request.

Workshop question Can it find scientifically meaningful candidates faster and more reliably than current workflows?

Evidence of progress Positive controls, rare-class retrieval, synthetic anomalies, baseline comparisons, and expert review.

2 Generative inference across instruments

Instruments see the universe through different resolutions, cadences, noise, coverage, and selection effects. Conditional generation may connect those views through super-resolution, deconfusion, gap filling, missing-modality prediction, or cross-instrument reconstruction.

Workshop question Is the generated view supported by evidence, rather than merely plausible-looking?

Evidence of progress Controlled degradation, held-out modalities, calibrated uncertainty, forward-model checks, and classical baselines.

Scientific standard: useful generative inference must expose what is observed, what is assumed, what remains uncertain, and how the output can be checked.

KEY PROBLEM AREAS

KPAs 3 and 4

3 Observations to physical states

Multimodal observations could constrain candidate physical states or simulation initial conditions. Candidate states are evolved through system and instrument simulators, then compared back to observations.

Workshop question Do forward runs remain physically admissible and observationally consistent?

Evidence of progress Forward-model consistency, stable physical structure, residual analysis, uncertainty, and an auditable chain of assumptions.

4 Human-machine discovery

This is the workflow lens: how should scientists, reasoning language models, tools, simulators, archives, and the joint-representation model form, test, revise, and communicate hypotheses together?

Workshop question Does the combined workflow improve search, triage, hypothesis generation, validation, or communication without hiding uncertainty or responsibility?

THE RELEVANCY GRAPH

Data become useful together when we know *why* they belong together: the same source, sky region, or time window; a simulation-to-observation forward model; an instrument response; shared code provenance; or workflow lineage. That graph connects scientific intent to model design and model outputs back to accountability.

BEFORE LEIDEN

Bring ingredients, not finished answers

ASTROPHYSICS + COMPUTATIONAL PHYSICS

Think about use cases where AI could help humans study the cosmos and do better science if it could understand, reason about, and generate physical data. Prepare a list of astrophysical data you can access. Any modality is welcome, including:

- observations, catalogues, spectra, images, time series, and event lists;
- simulation code, snapshots, initial conditions, and synthetic observations;
- multimodal observations of the same systems;
- instrument responses, calibration context, uncertainties, and provenance.

AI + COMPUTATIONAL INFRASTRUCTURE

Think about the architectural challenges facing a joint-representation model that must learn deep, self-supervised representations of large physical datasets. Also consider the infrastructure required to train across physical datasets that can dwarf the data used to train language models:

- long context, heterogeneous tokenisation, and shared latent spaces;
- compute-to-data training across distributed archives and HPC systems;
- data access, preprocessing, provenance, monitoring, and validation;
- staged scaling, simulator-backed synthetic data, and failure modes.

If you span both groups, bring both perspectives. If you span neither exactly, bring the constraints, examples, and questions from your own domain.

THE WEEK

One chain of daily deliverables

Day	Focus	Main deliverable
Monday	Vocabulary, use cases, evaluation	Shared Vocabulary and Discovery-Evaluation Map v0
Tuesday	Data, representation, provenance	Relevancy Graph v0 for selected use cases
Wednesday	Architecture and training	Architecture and Training Protocol v0
Thursday	MVP and hackathon	MVP Implementation Plan and Hackathon Artifacts v0
Friday	Roadmap, governance, funding	Consortium Roadmap and Funding Package v0

THE HAND-OFF

Monday chooses what matters. Tuesday defines the relationships the data and system must respect. Wednesday turns those constraints into a model and training path. Thursday turns that design into workstreams and artifacts. Friday gives the work owners, resources, governance, and continuity.

Working rhythm Each day combines brief framing talks, small-group work, plenary synthesis, and a closeout that confirms the artifact and its owners. Coffee breaks and lunches are part of the programme: use them to connect ideas across disciplines.

Formal working hours are 10:00–16:30 on Monday, 9:30–17:00 Tuesday–Thursday, and 9:30–13:30 on Friday.

MONDAY · 10:00–16:30

Use cases and discovery workflows

Deliverable Shared Vocabulary and Discovery-Evaluation Map v0

Time	Session
10:00–10:30	Kickoff talk Shared vocabulary for AI-accelerated discovery.
10:30–11:00	Coffee break
11:00–12:00	Use-case working session Draft cards covering workflow, roles, data, capability, and evidence of success.
12:00–13:30	Lunch
13:30–14:00	Afternoon talk Frame KPAs 1–4 as testable capabilities.
14:00–14:30	Refine use cases Choose mechanisms and a small set of flagship cases.
14:30–15:00	Coffee break
15:00–16:15	Define evaluation targets Controls, baselines, anomaly tests, forward checks, uncertainty, provenance, and human review.
16:15–16:30	Closeout Confirm vocabulary, cases, evaluations, and artifact owners.
16:30 onward	Welcome reception Informal continuation after the formal programme.

Keep asking What would count as evidence that this workflow produces better science, rather than simply more output?

TUESDAY · 9:30–17:00

Data, representation, and provenance

Deliverable Relevancy Graph v0 for Monday’s selected use cases

Time	Session
9:30–10:00	Kickoff talk The training corpus as a relevancy graph.
10:00–10:30	Map data objects Connect use cases to observations, simulations, instruments, code, and provenance.
10:30–11:00	Coffee break
11:00–12:00	Define the graph Agree node types, edge types, metadata, provenance, and ambiguities.
12:00–13:30	Lunch
13:30–14:00	Afternoon talk Tokenising heterogeneous physical data.
14:00–14:30	Tokenisation implications Units, scale, locality, order, uncertainty, missingness, modality, and context.
14:30–15:00	Coffee break
15:00–16:30	Plenary synthesis Build Relevancy Graph v0 and extract architecture constraints.
16:30–17:00	Closeout Confirm owners and open decisions.

Keep asking What relationship makes these objects scientifically relevant to one another, and what provenance must survive every transformation?

WEDNESDAY · 9:30–17:00

Long-context architecture and training

Deliverable Architecture and Training Protocol v0

Time	Session
9:30–10:00	Kickoff talk Long-context architectures for multimodal physics.
10:00–10:30	Translate graph constraints Preserve typed objects, relationships, units, uncertainty, and metadata.
10:30–11:00	Coffee break
11:00–12:00	Sketch architecture families Encoders, shared representations, retrieval, generative heads, simulators, and tool interfaces.
12:00–13:30	Lunch
13:30–14:00	Afternoon talk Training protocol and compute-to-data deployment.
14:00–14:30	Define training objectives Alignment, reconstruction, prediction, generation, forward consistency, and uncertainty.
14:30–15:00	Coffee break
15:00–16:30	Stage the training path Seed corpus, synthetic data, access, preprocessing, validation, scaling, and risks.
16:30–17:00	Closeout Confirm the architecture, protocol, open decisions, and owners.
17:00–20:00	Workshop dinner Informal continuation after the formal programme.

THURSDAY · 9:30–17:00

MVP architecture and hackathon

Deliverable MVP Implementation Plan and Hackathon Artifacts v0

Time	Session
9:30–10:00	Kickoff talk Turn the week's artifacts into MVP workstreams.
10:00–10:30	Define MVP scope Select the smallest credible system, graph, data, architecture, and evaluations.
10:30–11:00	Coffee break
11:00–12:00	Workstreams and backlog Name owners; write issues with acceptance criteria and links to use cases.
12:00–13:30	Lunch
13:30–14:00	Hackathon checkpoint Set the integration target across model, data, and evaluation.
14:00–14:30	Hackathon sprint 1 Schemas, manifests, examples, evaluation specs, prototypes, or CI.
14:30–15:00	Coffee break
15:00–16:00	Hackathon sprint 2 Continue implementation and integration.
16:00–16:30	Integration review Report repository artifacts, gaps, and next owners.
16:30–17:00	Closeout Finalise the plan, backlog, owners, and next technical meeting.

Thursday rule: every artifact should connect to a use case, a model capability, and an evaluation target.

FRIDAY · 9:30–13:30

Roadmap, governance, and funding

Deliverable Consortium Roadmap and Funding Package v0

Time	Session
9:30–10:00	Kickoff talk Compute, data access, and funding an open foundation-model effort.
10:00–10:30	Map resource needs Compute, data, simulators, archives, expertise, funding, and missing roles.
10:30–11:00	Coffee break
11:00–11:30	Funding and partnerships Identify opportunities, partners, institutional support, and lead owners.
11:30–12:00	Roadmap closeout Confirm milestones, working groups, owners, cadence, and invitees.
12:00–13:30	Closing lunch Final confirmations and departures.

PRACTICAL MATTERS

Discounted hotel reservations are limited, so please book as soon as possible if you need accommodation.

For practical questions or help with accommodation, contact Maria Krebbers at acceleratingastrophysical@lorentzcenter.nl.

accelerating-astrophysical-discovery.github.io

Bring a use case. Bring a constraint. Leave with something we can build together.